CSE STUDY HUB

Generative AI & Prompt Engineering

UNIT 3 · Generative AI Tools and Prompt Engineering

What this unit is about

This unit builds understanding of **Generative AI Tools and Prompt Engineering**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Generative tools for text, image, audio, video and code

2. Prompt anatomy: role, task, context, constraints and output format

3. Zero-shot, one-shot and few-shot prompting

4. Clear instructions, examples, delimiters and structured outputs

5. Prompt chaining and decomposition of complex tasks

6. Iterative refinement: test, inspect, revise and verify

7. Grounding with trusted context and retrieval-augmented generation concept

8. Prompt evaluation: accuracy, completeness, consistency and safety

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 3 NOTES · PART 1

Generative tools for text, image, audio, video and code

Generative tools for text, image, audio, video and code is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Prompt anatomy: role, task, context, constraints and output format

A strong prompt specifies the task, context, constraints and desired output format. Add examples when useful, break complex tasks into steps, and verify facts and assumptions in the response.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Zero-shot, one-shot and few-shot prompting

A strong prompt specifies the task, context, constraints and desired output format. Add examples when useful, break complex tasks into steps, and verify facts and assumptions in the response.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Generative AI Tools and Prompt Engineering, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 2

Clear instructions, examples, delimiters and structured outputs

Clear instructions, examples, delimiters and structured outputs is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Prompt chaining and decomposition of complex tasks

The OSI model has Physical, Data Link, Network, Transport, Session, Presentation and Application layers. Each layer provides services to the one above and uses the one below; encapsulation adds headers as data moves down.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Iterative refinement: test, inspect, revise and verify

Iterative refinement: test, inspect, revise and verify is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Generative AI Tools and Prompt Engineering, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 3

Grounding with trusted context and retrieval-augmented generation concept

Grounding with trusted context and retrieval-augmented generation concept is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Prompt evaluation: accuracy, completeness, consistency and safety

A strong prompt specifies the task, context, constraints and desired output format. Add examples when useful, break complex tasks into steps, and verify facts and assumptions in the response.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Generative AI Tools and Prompt Engineering, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Generative tools for text, image, audio, video and code** — know the definition, core working and one example.

• **Prompt anatomy: role, task, context, constraints and output format** — know the definition, core working and one example.

• **Zero-shot, one-shot and few-shot prompting** — know the definition, core working and one example.

• **Clear instructions, examples, delimiters and structured outputs** — know the definition, core working and one example.

• **Prompt chaining and decomposition of complex tasks** — know the definition, core working and one example.

• **Iterative refinement: test, inspect, revise and verify** — know the definition, core working and one example.

• **Grounding with trusted context and retrieval-augmented generation concept** — know the definition, core working and one example.

• **Prompt evaluation: accuracy, completeness, consistency and safety** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Generative AI Tools and Prompt Engineering. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Generative tools for text, image, audio, video and code and explain its importance.

Q2. Explain the working/steps involved in Prompt anatomy: role, task, context, constraints and output format.

Q3. Differentiate between two related concepts from Generative AI Tools and Prompt Engineering.

Q4. Give one practical application of Prompt evaluation: accuracy, completeness, consistency and safety and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Generative AI Tools and Prompt Engineering.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.